

ANTENNA AND WAVE PROPAGATION

Comprehensive Assignment-cum-Tutorial Sheet

Unit: Antenna Basics — 30 Questions

Scope note: This sheet is prepared strictly from the topics visible in the supplied textbook pages: radiation pattern and radiation intensity; beam solid angle; radian and steradian; directivity; gain; antenna/radiation efficiency; principle planes and beamwidth; HPBW and BWFN; bandwidth and Q; front-to-back ratio; antenna impedance and radiation resistance; self and mutual impedance; effective length/height; effective aperture; antenna/equivalent noise temperature; SNR; antenna polarization.

Instructions to Students

- Attempt all 30 questions. Show the formula, substitution, units and final answer for numerical problems.
- For pattern-related questions, clearly state the angular limits and assumptions used.
- Use consistent SI units. Give directivity/gain in linear form and dB when requested.
- Use the hints only as guidance; do not treat them as complete solutions.
- For case studies, state assumptions and justify conclusions using the antenna parameters named in the question.

Section A — Conceptual and Tutorial Questions (Q1–Q24)

1. Define field radiation pattern, power radiation pattern, radiation intensity, and beam solid angle. State how these quantities are related to the directional behavior of an antenna. [Bloom: Remember | CO: CO1 | Marks: 2]
2. Explain the difference between a radian and a steradian. Give the angular extent represented by 2π steradians and 4π steradians, and state their significance in antenna radiation calculations. [Bloom: Understand | CO: CO1 | Marks: 3]
3. Explain radiation intensity $U(\theta, \phi)$ and distinguish it from power density at a point. State the physical meaning of maximum radiation intensity U_0 . [Bloom: Understand | CO: CO1 | Marks: 3]
4. Define directivity D of an antenna and derive its relation with maximum radiation intensity and total radiated power. [Bloom: Understand | CO: CO1 | Marks: 5]
5. Define antenna gain G and antenna efficiency η . Derive the relationship between gain, directivity and efficiency. [Bloom: Understand | CO: CO1 | Marks: 5]
6. State the condition under which antenna gain and directivity are equal. Explain the physical meaning of that condition. [Bloom: Understand | CO: CO1 | Marks: 2]
7. An antenna has radiation resistance 73Ω and loss resistance 7Ω . If its power gain is 20 (linear), calculate antenna efficiency and directivity. Also express directivity in dB. [Bloom: Apply | CO: CO2 | Marks: 5]
8. An antenna has radiation resistance 72Ω , loss resistance 8Ω and power gain 12 dB. Calculate antenna efficiency and directivity in dB. [Bloom: Apply | CO: CO2 | Marks: 5]
9. A voltage source of $(100 + j80) \text{ V}$ with source impedance 50Ω feeds an antenna whose impedance is $(71 + j25) \Omega$ including a 1Ω loss resistance. Calculate the radiation efficiency, real power delivered by the source, real power input to the antenna, power radiated, and power dissipated in the antenna loss resistance. [Bloom: Apply | CO: CO2 | Marks: 8]
10. Define the principal (principle) planes of an antenna radiation pattern. Explain why principal-plane patterns are useful when describing antenna beamwidth. [Bloom: Understand | CO: CO3 | Marks: 3]

11. Explain the method of beamwidth measurement from an antenna radiation pattern. Define half-power points and describe how HPBW is obtained. [Bloom: Understand | CO: CO3 | Marks: 4]
12. Differentiate HPBW and BWFN. Explain what information each beamwidth gives about an antenna pattern. [Bloom: Understand | CO: CO3 | Marks: 3]
13. Define front-to-back ratio (FBR). Explain how it is obtained from a radiation pattern and why it is useful for a directional antenna. [Bloom: Understand | CO: CO3 | Marks: 3]
14. A unidirectional source has radiation intensity $U(\theta, \phi) = U_0 \sin\theta \sin^2\phi$ over $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq \pi$. Set up the expression for total radiated power and the corresponding directivity. Do not skip the angular integration limits. [Bloom: Apply | CO: CO3 | Marks: 6]
15. For a unidirectional source with radiation intensity $U = 6 \cos\theta$ over its stated radiation region, determine the directivity. Clearly identify the maximum radiation intensity and the total radiated power used. [Bloom: Apply | CO: CO2 | Marks: 5]
16. An antenna has a bandwidth of 600 kHz and is centered at 30 MHz. Calculate its Q. State the bandwidth relation used. [Bloom: Apply | CO: CO2 | Marks: 3]
17. Define antenna bandwidth. Explain why bandwidth is an important antenna parameter and distinguish it from beamwidth. [Bloom: Understand | CO: CO1 | Marks: 3]
18. Define antenna impedance and radiation resistance. Explain the role of loss resistance in the total antenna resistance. [Bloom: Understand | CO: CO4 | Marks: 4]
19. An antenna impedance is represented as $R + jX$. Explain the physical meaning of the resistive and reactive parts. State what portion of the resistance is associated with radiation and what portion represents loss. [Bloom: Understand | CO: CO4 | Marks: 3]
20. Define self impedance and mutual impedance of antennas. Explain how mutual impedance is relevant when more than one antenna element is present. [Bloom: Understand | CO: CO4 | Marks: 4]
21. Define effective length (effective height) of an antenna. Explain its significance in relating an incident electric field to the received voltage. [Bloom: Understand | CO: CO4 | Marks: 4]
22. Define effective aperture of an antenna and derive its relationship with antenna gain and wavelength. [Bloom: Apply | CO: CO4 | Marks: 5]
23. Define antenna temperature and equivalent noise temperature. Explain the meaning of signal-to-noise ratio (SNR) in an antenna receiving system. [Bloom: Understand | CO: CO4 | Marks: 4]
24. Write a short note on antenna polarization. Explain how polarization is described and why polarization should be considered when transmitting and receiving a signal. [Bloom: Understand | CO: CO4 | Marks: 4]

Section B — Case Studies and Engineering Application (Q25–Q30)

Case-study instruction: For each case, identify the relevant antenna parameter(s), write the governing relationship, perform the required calculation or comparison, and justify the conclusion.

25. Case Study 1 — Efficiency versus directivity: Two antennas are used at the same operating condition. Antenna A has radiation resistance 73Ω , loss resistance 7Ω and power gain 20 (linear). Antenna B has radiation resistance 72Ω , loss resistance 8Ω and power gain 12 dB. Calculate η and D for both and compare which antenna has the higher directivity and which has the higher efficiency. [Bloom: Analyze | CO: CO5 | Marks: 8]
26. Case Study 2 — Feed network and power flow: A source of $(100 + j80)$ V with 50Ω impedance feeds an antenna of $(71 + j25) \Omega$. The antenna includes 1Ω loss resistance. Using the same model as the numerical example, determine the power flow from source to antenna, the radiated power and the dissipated loss power. Comment on where the input power is distributed. [Bloom: Analyze | CO: CO5 | Marks: 8]
27. Case Study 3 — Pattern selection: An antenna designer needs a directional pattern and is comparing two measured radiation patterns. Pattern A has a narrow main lobe with a clearly measurable half-power point; Pattern B has a wider main lobe but lower response in the rear direction. Identify which beamwidth parameter

should be extracted for main-lobe width, which parameter describes front-versus-back response, and what additional pattern information is needed before declaring one antenna superior. [Bloom: Evaluate | CO: CO5 | Marks: 6]

28. Case Study 4 — Aperture-based gain: A circular-aperture antenna has diameter 1.22 m at 6 GHz. Its maximum effective aperture is 55% of its physical aperture. Calculate the antenna gain from the effective-aperture relationship and express the result in dB. State each intermediate quantity used. [Bloom: Apply | CO: CO2 | Marks: 7]

29. Case Study 5 — Aperture size and gain: A circular-aperture antenna has diameter 3 m and operates at 5 GHz with zero losses. Calculate its gain. Then explain, using the aperture/gain relationship, why the physical aperture size and wavelength affect the gain. [Bloom: Analyze | CO: CO5 | Marks: 7]

30. Case Study 6 — Receiving-system assessment: A receiving antenna is characterized by effective aperture, antenna/equivalent noise temperature and SNR. Prepare a short engineering assessment stating (i) what effective aperture indicates, (ii) what antenna temperature represents, (iii) how equivalent noise temperature is used in a noise model, and (iv) why SNR is a useful receiving-system measure. Conclude by identifying which quantities are geometric/radiation-related and which are noise-related. [Bloom: Evaluate | CO: CO5 | Marks: 8]

Hints / Guidance

- Q1:** Use the standard definitions and distinguish field amplitude from power/radiation quantities.
- Q2:** A radian is a plane angle; a steradian is a solid angle. A complete sphere subtends 4π sr.
- Q3:** Radiation intensity is power per unit solid angle. Its maximum is U_0 .
- Q4:** Start with $D = U_{\max} / U_{\text{avg}}$ and $U_{\text{avg}} = \text{Prad} / (4\pi)$.
- Q5:** Use $G = \eta D$, where η is the radiation/antenna efficiency used in the given examples.
- Q6:** Set $\eta = 1$ in $G = \eta D$.
- Q7:** $\eta = R_r / (R_r + R_{\text{loss}})$; then $D = G/\eta$ and convert D to dB with $10 \log_{10}(D)$.
- Q8:** Convert 12 dB to linear gain before using $D = G/\eta$.
- Q9:** Find the circuit current from the source and total series impedance. Use real power expressions and separate radiation and loss resistance.
- Q10:** Principal planes are selected angular cuts through the 3-D pattern; use them to report directional characteristics in a convenient 2-D form.
- Q11:** Locate the points at which power is one-half of maximum (-3 dB) and measure the angular separation.
- Q12:** HPBW uses half-power points; BWFN uses the first-null points.
- Q13:** Compare radiation intensity/power pattern in the desired direction with that in the rear direction and express the ratio in dB.
- Q14:** Use $\text{Prad} = \iint U(\theta, \phi) \sin\theta \, d\theta \, d\phi$ over the stated region, then $D = 4\pi U_0 / \text{Prad}$.
- Q15:** Because the source is unidirectional, use the radiation region where U is non-negative and integrate $U \sin\theta$ over that region.
- Q16:** Use $Q = f_0 / \text{BW}$ with consistent frequency units.
- Q17:** Bandwidth is the frequency range satisfying the stated performance criterion; beamwidth is an angular measure of a radiation pattern.
- Q18:** For a simple antenna model, total resistance is the sum of radiation and loss resistances.
- Q19:** R accounts for real-power transfer; X represents reactive behavior. Split R into radiation and loss parts when specified.
- Q20:** Self impedance describes an element's own voltage-current relationship; mutual impedance describes coupling between elements.
- Q21:** Relate received open-circuit voltage to incident electric field through effective length.
- Q22:** Use $A_e = \lambda^2 G / (4\pi)$.
- Q23:** Antenna temperature is a temperature-equivalent representation of received noise power; equivalent noise temperature represents noise in temperature form; SNR is signal power relative to noise power.
- Q24:** State the orientation/relationship of the electric-field vector and the antenna's polarization, and note the importance of polarization matching.
- Q25:** Compute each efficiency first, then directivity. Compare like quantities separately.
- Q26:** Use the source/antenna equivalent circuit. The $1 \, \Omega$ term is the specified loss resistance.
- Q27:** Main-lobe width $\rightarrow$ HPBW; front/rear response $\rightarrow$ FBR. A complete comparison also needs relevant pattern, gain/directivity, efficiency, and measurement conditions.
- Q28:** Physical area of a circle is $\pi d^2/4$; effective aperture = 0.55 times physical aperture; $\lambda = c/f$; then use $A_e = \lambda^2 G / (4\pi)$.
- Q29:** Use the circular physical aperture $A = \pi d^2/4$ and zero-loss gain relationship. Keep frequency in Hz when calculating wavelength.
- Q30:** Separate receiving aperture/collection concepts from noise-temperature and SNR concepts; do not treat all four as the same parameter.

Formula Quick Reference for Tutorials

Formulae
$\eta = R_r / (R_r + R_{\text{loss}})$
$G = \eta D$
$D = U_{\max} / U_{\text{avg}} = 4\pi U_{\max} / \text{Prad}$
$\text{Prad} = \iint U(\theta, \phi) \, d\Omega = \iint U(\theta, \phi) \sin\theta \, d\theta \, d\phi$

$Q = f_0 / BW$
$A_e = \lambda^2 G / (4\pi)$
Physical area of a circular aperture: $A = \pi d^2/4$
$\lambda = c/f$
Power-ratio in dB: $10 \log_{10}(P_2/P_1)$

Bloom's Taxonomy Legend

Remember: recall/define. **Understand:** explain/distinguish. **Apply:** calculate/use relationships. **Analyze:** interpret/compare/break into relationships. **Evaluate:** justify/select/conclude.

Assessment Coverage Summary

Coverage area	Question numbers	Primary CO	Typical Bloom level
Fundamentals, angles, radiation intensity	1–4	CO1	Remember–Understand
Gain, directivity, efficiency and numerical analysis	5–9, 15–16, 25	CO2	Apply–Analyze
Pattern, principal planes, beamwidth and FBR	10–14, 27	CO3	Understand–Evaluate
Impedance, aperture, effective length, noise, SNR, polarization	18–24, 28–30	CO4/CO5	Understand–Evaluate
Integrated engineering cases	25–30	CO5	Analyze–Evaluate